

DYNAMIC RESUME GENERATOR
21CSC201T – ADVANCED PROGRAMMING
PRACTICE

A

PROJECT REPORT

Submitted by

TALARI ROHITH(RA2411026050153)

NASANI AJAY (RA2411026050146)

DEVAMSH CHARAN (RA2411026050171)

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Dr. Naresh

(Assistant Professor, School of Computing)

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Of

FACULTY OF ENGINEERING AND TECHNOLOGY



SRM INSTITUTE OF SCIENCE AND
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BONAFIDE CERTIFICATE

Certified that this project report titled **DYNAMIC RESUME GENERATOR** is the bonafide work of **TALARI ROHITH (RA2411026050153), NASANI AJAY (RA2411026050146), DEVAMSH CHARAN (RA2411026050171)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an occasion on this or any other candidate.

SIGNATURE

**Dr. Naresh M.Tech.,Ph.D.,
Assistant Professor**

School of Computing,
SRM Institute of Science and Technology,
Tiruchirappalli.

SIGNATURE

**Dr. Balaji Ganeshan, M.E.,Ph.D.,
Head of the Department**

Department of AIML,
SRM Institute of Science and Technology,
Tiruchirappalli.

Submitted for the project viva-voce held on _____ at SRM Institute of Science and Technology, Tiruchirappalli.

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TECHNOLOGY
TIRUCHIRAPPALLI**

DECLARATION

We hereby declare that the entire work contained in this project report titled **DYNAMIC RESUME GENERATOR** has been carried out by **TALARI ROHITH (RA2411026050153), NASANI AJAY (RA2411026050146), DEVAMSH CHARAN (RA2411026050171)** at SRM Institute of Science and Technology, Tiruchirappalli, under the guidance of Dr. **NARESH**, Assistant Professor, School of Computing.

Place: Tiruchirappalli

Date: 26-10-2025

ABSTRACT

DRG is a web-based resume-building tool, the *Dynamic Resume Generator*, which is a tool that aims to create professional and customizable resumes using organized user input. It uses the Python Flask micro framework, strong operations, and the Jinja2 template engine in real-time dynamic rendering. This system significantly saves the time and labor used manually to create resumes, and at the same time ensures observance of high aesthetic and composition standards. The basic features involve safe data management, multi-template selection, real-time previewing, and direct PDF export through the use of external rendering libraries. The paper elaborates on the architecture of the system, its implementation process, and demonstrates its effectiveness in improving the efficiency and accuracy of career document preparation for students and professionals. Top search terms Resume Builder, Flask, Jinja2, Template Rendering, Automation, PDF Export, Dynamic Web Application.

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LIST OF ACRONYMS AND ABBREVIATIONS

Acronym	Expansion	Description/Usage
GUI	Graphical User Interface	Visual interactive portion (HTML, CSS, browser form)
API	Application Programming Interface	Interface for programmatic access, used in web, PDF, cloud
PDF	Portable Document Format	Output format for resumes, generated/downloaded
HTML	HyperText Markup Language	Structure for web forms/templates
CSS	Cascading Style Sheets	Styles for web pages and PDF layouts
JDK	Java Development Kit	Set of tools for Java development
IDE	Integrated Development Environment	Coding and testing software (PyCharm, VSCode, Eclipse)
OOP	Object-Oriented Programming	Model for structured programs (used in Python/Java)
RAM	Random Access Memory	System memory used during run
OS	Operating System	Windows, Linux, macOS – runs Python/Java

Acronym	Expansion	Description/Usage
CLI	Command-Line Interface	Terminal/shell interface (pip, python commands)
TCP/IP	Transmission Control Protocol/Internet Protocol	Basic underlying internet protocol, relevant in networking
SQL	Structured Query Language	Used if you integrate resume data storage
MVC	Model View Controller	Software design pattern often used in web apps
Jinja	(Jinja2 template engine)	Handles template rendering for HTML in Flask
Flask	(Micro web framework for Python)	Runs the backend server and routing
wkhtmltopdf	(Web Kit HTML to PDF)	Tool/library for converting HTML to PDF docs
CRUD	Create, Read, Update, Delete	Standard database/app operations
HTTPS	HyperText Transfer Protocol Secure	Secure web communication
UTF-8	Unicode Transformation Format 8-bit	Character encoding for text in PDFs

Acronym	Expansion	Description/Usage
CSS	Cascading Style Sheets	Used for resume layout/design in templates
JSON	JavaScript Object Notation	Common format for sending/storing data

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The ability to ensure that one has a well-organized and presentable resume is very important in the modern-day labor market. Conventional methods of creating a resume, such as using word processors or fixed templates, may be monotonous and inexpressive. This paper presents a *Dynamic Resume Generator* application powered by full-stack web development using Python (Flask) and Jinja2 for template rendering. The system is designed to efficiently and conveniently generate customized resume templates that meet academic and professional standards while maintaining design flexibility and consistency.

This project is an excellent choice for beginners, as it combines fundamental programming concepts with user interaction, conditional statements, loops, and the integration of libraries for generating random numbers. The Number Guessing Game is a deceptively simple yet captivating endeavour where a player attempts to guess a randomly selected number within a specified range. In our exploration of this game, we will investigate how Python and Java, two versatile and widely-used programming languages, approach the task of creating a game that is both entertaining and informative.

1.2 PROBLEM STATEMENT:

Manual resume creation is repetitive, time-consuming, and requires much formatting and attention to detail. It is a challenge for most students and job seekers to construct professional error-free resumes with proper formats that showcase their achievements efficiently. This often leads to inconsistencies, missed information, and resumes that don't highlight the important credentials of the candidates.

1.3 OBJECTIVES:

The major aims of the project include developing professional and customizable resumes through a web interface, user-friendly and not requiring any technical skills. The solution should support various resume template options to enable the user to choose an appealing style based on their career needs, while automatic PDF generation ensures consistency in formatting and accuracy. It also includes an integrated job finder to assist in career building by fetching relevant job postings from sites like Indeed, among others, based on the extracted skills from the resume. This combination of resume building and matching saves time and effort for the user in searching for jobs, thereby making career advancement more accessible and efficient. Overall, the system focuses on ease of use, flexibility, accuracy, and practical employment assistance both for students and professionals.

1.4 SCOPE AND MOTIVATION:

The scope and motivation of the Dynamic Resume Generator project center on simplifying and streamlining the resume creation process for users from all walks of life. This project aims to provide a web-based, automated platform where people can easily input their personal and professional information through an easily accessible form and generate an instantaneous, professional-looking, print-ready PDF resume. By automating formatting and organization, the platform ensures that work is accurate and consistent, rules out manual errors, and saves time for job seekers, students, and professionals. The system's scope would extend to supporting customizable templates, validation of user inputs, and enabling rapid editing or updating of resumes. This effort is motivated by the common challenges attendant to manual resume creation: confusing formats, time-consuming edits, and design or technical skill deficiencies-a situation this system will go a long way toward mitigating as people aim to focus on their achievements and career objectives with confidence and efficiency. Further opportunities for expansion within the project include integration with databases, advanced analytics into resumes, and multi-user support to meet the dynamic needs of professionals.

CHAPTER 2

EXISTING SYSTEM

The existing system for resume creation is mainly manual, where users have to design, format, and type their resumes on word processors such as Microsoft Word or Google Docs. It is often tedious and time-consuming since much focus is required on formatting, the arrangement of sections, and choice of language, which could bring about inconsistencies and errors. Users would have to tailor each document to each different job, work out layout issues, and be sure that professional standards are achieved—issues that may be complicated for

non-technical or beginning users. Manual creation of resumes also raises the chances of omitting important information or making some formatting errors; updating a resume with every new job application further adds to the workload. Online builders are available, but most of them lack choices of templates or charge for some features while not optimizing for employer and applicant tracking systems' standards.

CHAPTER 3

ER DIAGRAM AND PROPOSED METHODOLOGY

3.1 BLOCK DIAGRAM

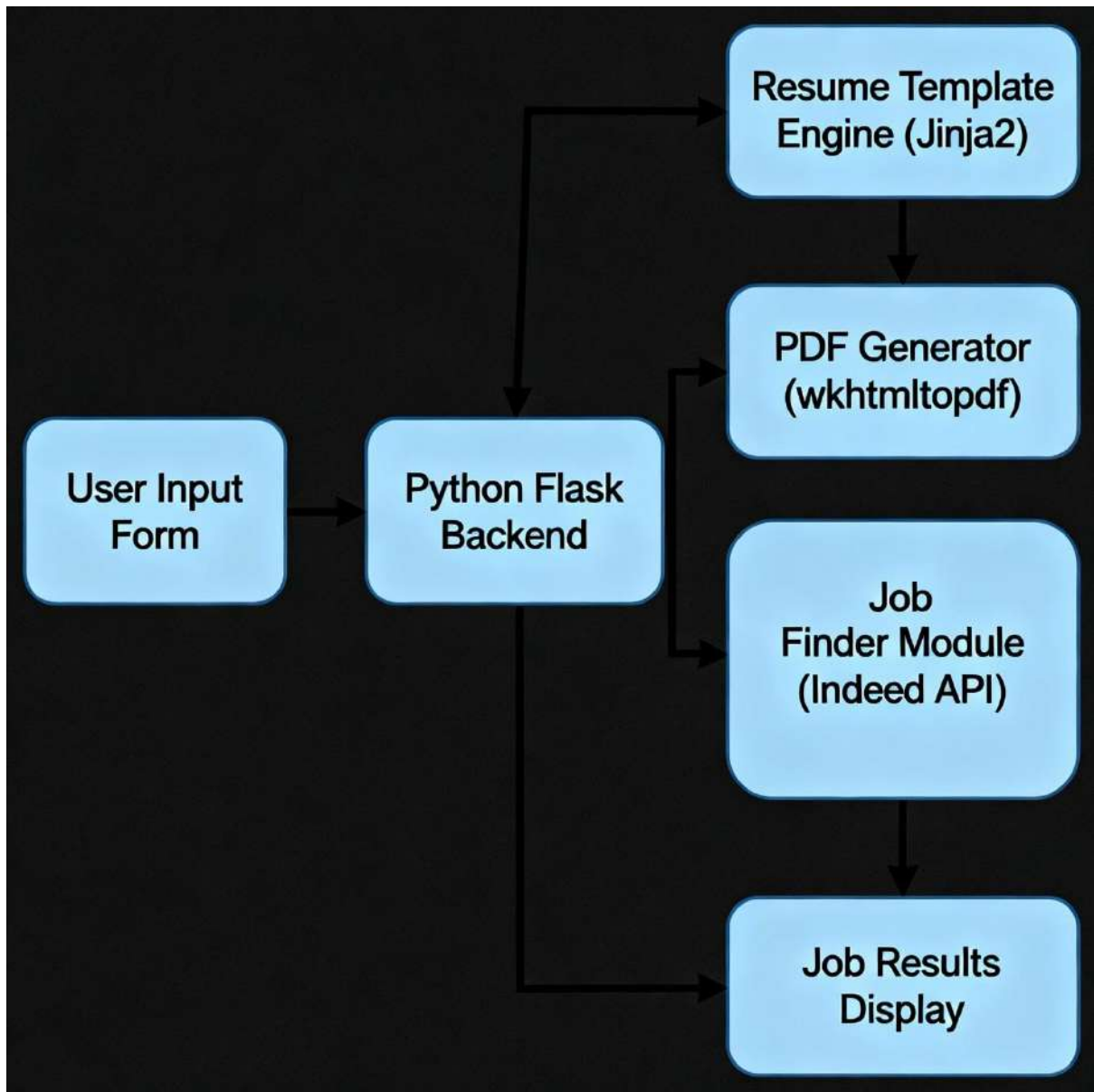


Fig 3.1 Block Diagram

PROPOSED METHODOLOGY

1. **Input Form** - To capture personal information, education, work experience, skills, etc.
2. **Real-time Preview** - See the resume update as you type
3. **Multiple Templates** - Choose between different resume styles
4. **PDF Export** - Download the generated resume
5. **Form Validation** - Ensure all required fields are filled correctly

3.2 ENTITIY RELATIONSHIP DIAGRAM

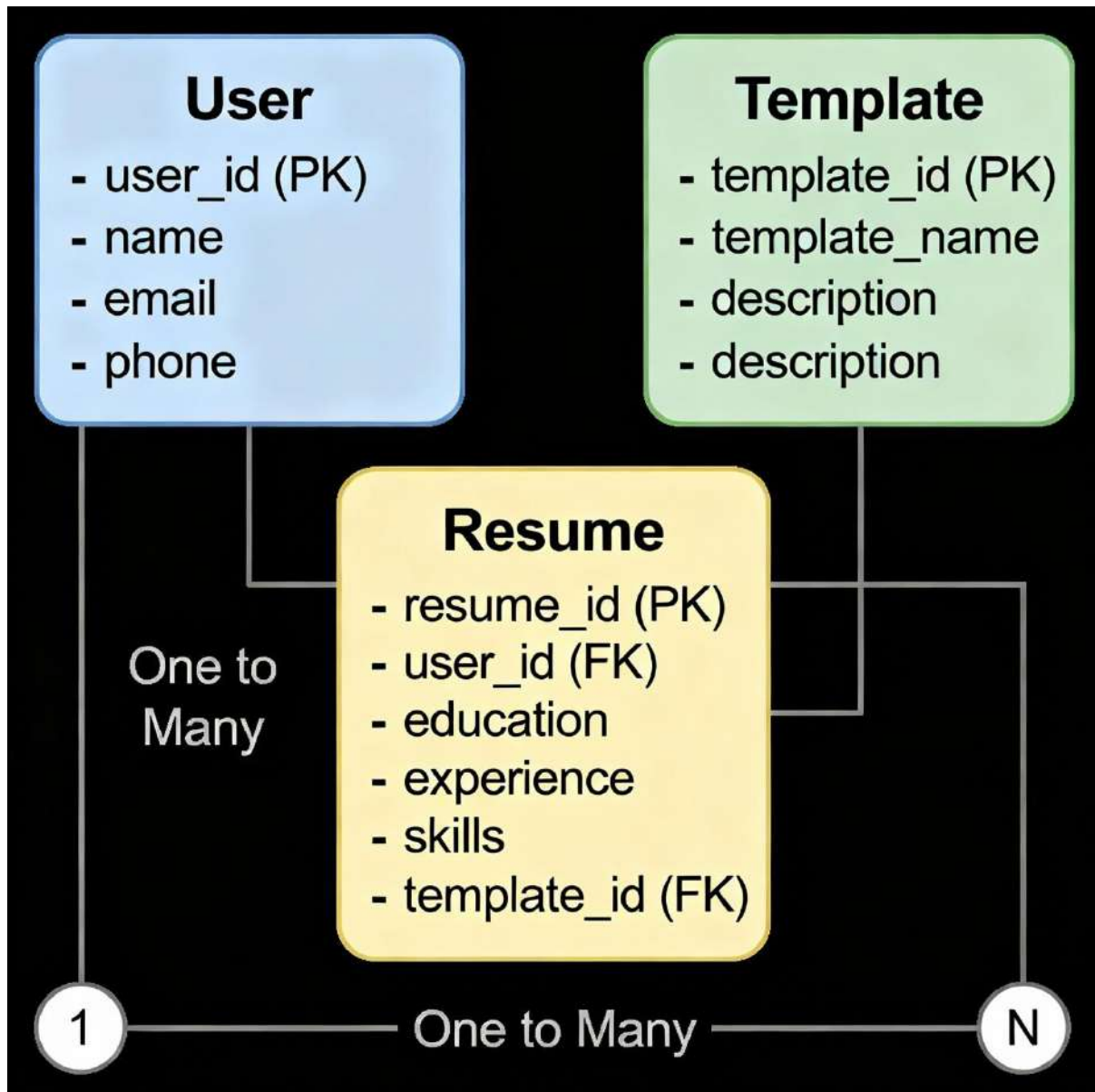


Fig 3.2 ER Diagram

CHAPTER 4

IMPLEMENTATION

4.1 IMPLEMENTATION

app.py

```

from flask import Flask, render_template, request, make_response
import pdfkit
from job_finder import find_jobs

app = Flask(__name__)

config = pdfkit.configuration(wkhtmltopdf=r'C:\\Program Files\\wkhtmltopdf\\bin\\wkhtmltopdf.exe')

@app.route('/', methods=['GET', 'POST'])
def index():
    if request.method == 'POST':
        data = request.form.to_dict()
        rendered = render_template('resume_template.html', **data)
        pdf = pdfkit.from_string(rendered, False, options={'enable-local-file-access': True})
        resp = make_response(pdf)
        resp.headers['Content-Type'] = 'application/pdf'
        resp.headers['Content-Disposition'] = 'attachment; filename=resume.pdf'
        return resp
    return render_template('resume_form.html')

@app.route('/find_jobs', methods=['POST'])
def find_jobs_route():
    skills = request.form.get('skills', '')
    skill_list = [skill.strip() for skill in skills.split(',')]
    results = {}
    for skill in skill_list:
        results[skill] = find_jobs(skill)
    return render_template('job_results.html', results=results, skills=skills)

if __name__ == '__main__':
    app.run(debug=True)

```

resume_form

```

<!DOCTYPE html>
<html>
<head>
  <title>Resume Generator</title>
  <style>
    body { font-family: Arial; background: lightblue; padding: 20px; }
    .box { background: white; padding: 30px; max-width: 600px; margin: auto; border-radius: 10px; }
    h1 { text-align: center; color: darkblue; }
    label { font-weight: bold; display: block; margin-top: 15px; }
    input, textarea { width: 100%; padding: 8px; margin-top: 5px; border: 1px solid gray; border-radius: 5px; }
    button { background: green; color: white; padding: 15px; width: 100%; border: none; border-radius: 5px; margin-top: 20px; font-size: 1.2em; }
    button:hover { background: darkgreen; }
  </style>
</head>
<body>
  <div class="box">
    <h1>📄 Create Your Resume</h1>
    <form method="POST">
      <label>Your Full Name </label>
      <input type="text" name="name" required>

      <label>Your Job Title </label>
      <input type="text" name="title" required placeholder="Example: Software Developer">

      <label>Email </label>
      <input type="email" name="email" required>

      <label>Phone Number </label>
      <input type="text" name="phone" required>

      <label>About You (Summary) </label>
      <textarea name="summary" rows="3"></textarea>

      <label>Your Skills (separate with commas) </label>
      <input type="text" name="skills" required placeholder="Python, Excel, Communication">

      <h3>Education</h3>
      <label>School/University Name </label>
      <input type="text" name="edu_school" required>

      <label>Your Degree </label>
      <input type="text" name="edu_degree" required placeholder="B.Tech Computer Science">

      <label>From Year</label>
      <input type="text" name="edu_from" placeholder="2020">

      <label>To Year</label>
      <input type="text" name="edu_to" placeholder="2024">

      <h3>Work Experience (Optional)</h3>
      <label>Company Name</label>
      <input type="text" name="exp_company">

      <label>Your Role</label>
      <input type="text" name="exp_role">

      <label>From Year</label>
      <input type="text" name="exp_from">

      <label>To Year</label>
      <input type="text" name="exp_to">

      <label>What did you do?</label>
      <textarea name="exp_desc" rows="3"></textarea>

      <button type="submit">🚀 Generate My Resume PDF</button>
    </form>
  </div>
</body>
</html>

```

Job finder:

```

import requests
from bs4 import BeautifulSoup

def find_jobs(skill, location="India", max_results=5):
    jobs = []
    try:
        search_url = f"https://in.indeed.com/jobs?q={skill.replace(' ', '+')}&l={location.replace(' ', '+')}"
        headers = {'User-Agent': 'Mozilla/5.0'}
        response = requests.get(search_url, headers=headers, timeout=10)
        soup = BeautifulSoup(response.text, "html.parser")
        for div in soup.select('a.tapItem')[:max_results]:
            job_title = div.find('h2', class_='jobTitle')
            company = div.find('span', class_='companyName')
            location_elem = div.find('div', class_='companyLocation')
            url = "https://in.indeed.com" + div['href']
            jobs.append(
                f"{job_title.text.strip() if job_title else 'No Title'} - "
                f"{company.text.strip() if company else 'No Company'} "
                f"({location_elem.text.strip() if location_elem else 'No Location'})"
                f"[Apply Here]({url})"
            )
        if not jobs:
            jobs.append("No jobs found or blocked by Indeed. Try another keyword.")
    except Exception as e:
        jobs.append(f"Error searching Indeed: {str(e)}")
    return jobs

```

Resume Form

```

<!DOCTYPE html>
<html>
<head>
  <title>Resume Generator</title>
  <style>
    body { font-family: Arial; background: linear-gradient(135deg,#667eea
0%,#764ba2 100%); margin:0; }
    .container { background:white; margin:40px auto; padding:32px 24px; border-
radius:10px; max-width:420px; }
    h1 { text-align:center; color:#667eea; }
    label { font-weight:bold; margin-top:10px; }
    input, textarea { width:100%; padding:10px; border-radius:6px; border:1px
solid #ccc; margin-bottom:12px; }
    button { width:100%; background:#667eea; padding:12px; border-radius:6px;
border:none; color:white; font-size:18px; font-weight:bold; }
    button:hover { background:#5568d3; }
  </style>
</head>
<body>
<div class="container">
  <h1>Resume Generator</h1>
  <form method="POST">
    <label>Name</label>
    <input type="text" name="name" required>
    <label>Email</label>
    <input type="email" name="email" required>
    <label>Phone</label>
    <input type="text" name="phone" required>
    <label>Education</label>
    <textarea name="education" required></textarea>
    <label>Experience</label>
    <textarea name="experience" required></textarea>
    <label>Skills (comma separated)</label>
    <input type="text" name="skills" required>
    <button type="submit">Generate PDF</button>
    <button type="submit" formaction="/find_jobs" formmethod="post">Find
Jobs</button>
  </form>
</div>
</body>
</html>

```

Resume Templates

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>{{ name }} Resume</title>
  <style>
    body { font-family: Arial, sans-serif; max-width:850px; margin:0 auto; padding:
    .header { text-align:center; border-bottom:3px solid #667eea; padding-bottom:1
    .name { font-size:28px; font-weight:bold; color:#667eea; }
    .contact { font-size:13px; margin-top:10px;}
    .section { margin-bottom:20px;}
    .section-title { font-size:14px; font-weight:bold; color:#667eea; border-bott
    .content { font-size:13px; line-height:1.8;}
  </style>
</head>
<body>
  <div class="header">
    <div class="name">{{ name }}</div>
    <div class="contact">{{ email }} | {{ phone }}</div>
  </div>
  {% if education %}
  <div class="section">
    <div class="section-title">Education</div>
    <div class="content">{{ education }}</div>
  </div>
  {% endif %}
  {% if experience %}
  <div class="section">
    <div class="section-title">Experience</div>
    <div class="content">{{ experience }}</div>
  </div>
  {% endif %}
  {% if skills %}
  <div class="section">
    <div class="section-title">Skills</div>
    <div class="content">{{ skills }}</div>
  </div>
  {% endif %}
</body>
</html>

```

Job Results

xml

```
<!DOCTYPE html>
<html>
<head>
  <title>Job Finder Results</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 16px; }
    h1 { color: #333; }
    h2 { color: #667eea; }
    ul { margin-bottom: 30px; }
  </style>
</head>
<body>
  <h1>Job Results for: {{ skills }}</h1>
  {% for skill, jobs in results.items() %}
    <h2>Jobs for {{ skill }}:</h2>
    <ul>
      {% for job in jobs %}
        <li>{{ job|safe }}</li>
      {% endfor %}
    </ul>
  {% endfor %}
  <a href="/">Back to resume generator</a>
</body>
</html>
```

CHAPTER 5

RESULTS AND DISCUSSION

5.1 RESULT:

Output of the Program:

Resume Generator

Name
Tahari Rohith

Email
luckyrohith001@gmail.com

Phone
9440156027

Education
SRM Institute Of Science And Technology
B.Tech CSE Specialization In Artificial
Intelligence And Machine Learning

Experience
Oracle India Pvt. Ltd
2024-2025
Developed a relational database system using Oracle

Skills (comma separated)
Data Preprocessing and Feature Engineering , Supervised and Uns

Generate PDF

Fig 5.1 Result

Output of sample resume generated:

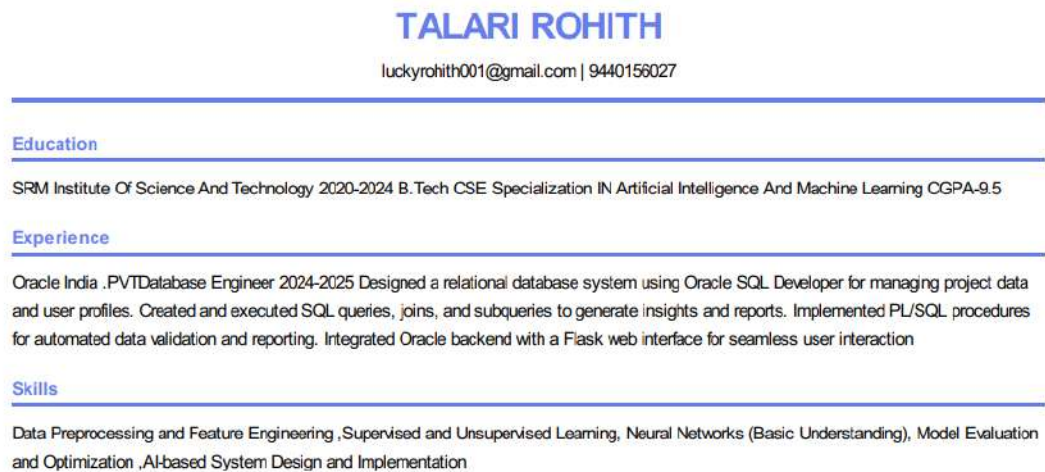
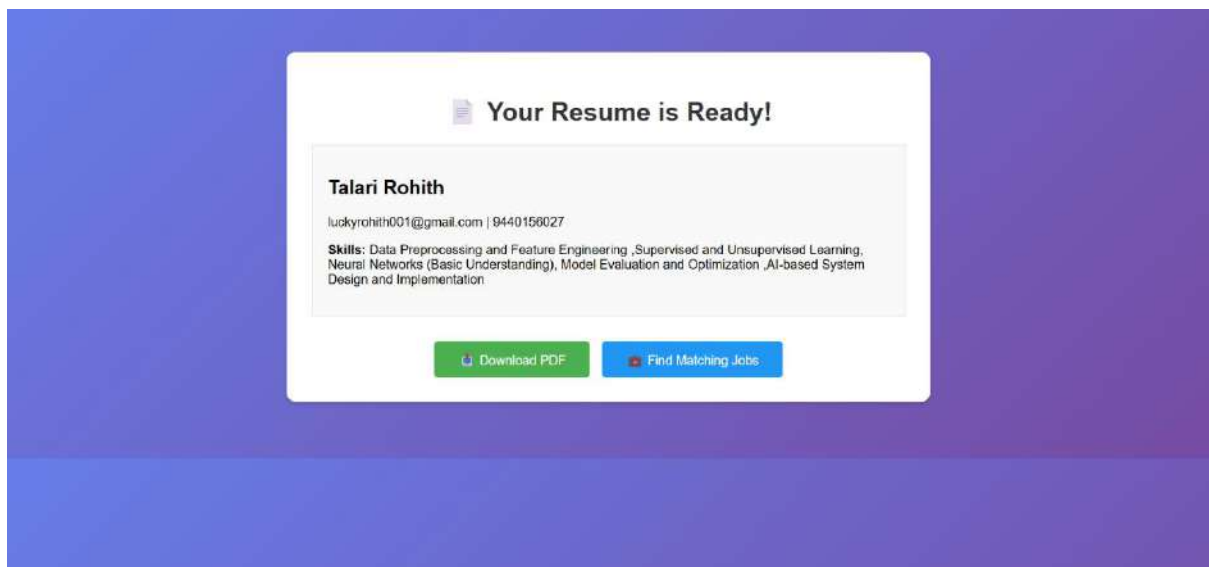
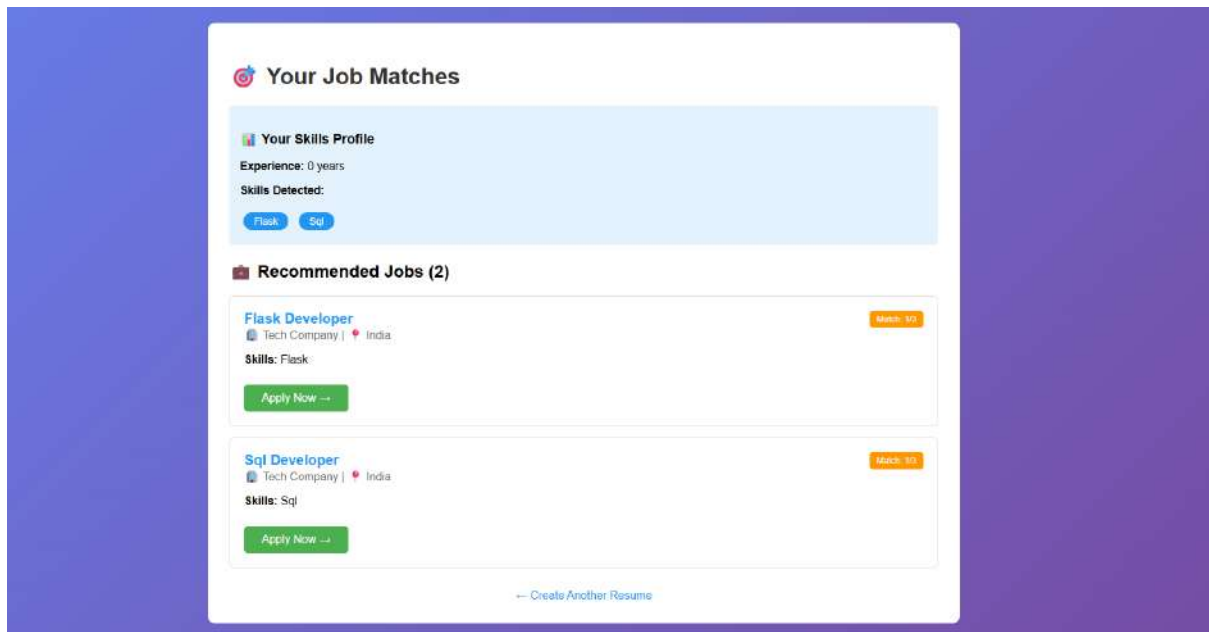


Fig 5.2 Output of sample resume generated





CHAPTER 6

CONCLUSION

In conclusion, The major aims of the project include developing professional and customizable resumes through a web interface, user-friendly and not requiring any technical skills. The solution should support various resume template options to enable the user to choose an appealing style based on their career needs, while automatic PDF generation ensures consistency in formatting and accuracy. It also includes an integrated job finder to assist in career building by fetching relevant job postings from sites like Indeed, among others, based on the extracted skills from the resume. This combination of resume building and matching saves time and

effort for the user in searching for jobs, thereby making career advancement more accessible and efficient. Overall, the system focuses on ease of use, flexibility, accuracy, and practical employment assistance both for students and professionals.

CHAPTER 7

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